1. Model Description

Parameter	Value
Model	AML_AE26_ChainDrive__04_HLA_Var.etd
Gap variants	0.0042 mm (nominal-tight) and 0.008 mm (loose, ~1.9× nominal)
Speed points	7400 / 7500 / 7600 / 7700 rpm
Intake elements	16 (INTr_VAFA1-8, INTL_VAFA1-8)
Exhaust elements	16 (EXHr_VAFA1-8, EXHL_VAFA1-8)
Analysis window	Last cam cycle (360°) of 5400° total (15 cycles)
Seat contact threshold	0.05 mm lift
Bounce threshold	0.2 mm re-lift after contact
Software	AVL EXCITE Timing Drive R2024.1
Simulation completed	2026-07-10
Report date	2026-07-21

This report evaluates **valve closing behaviour** — lift profiles, seat impact velocity, and bounce — across two HLA leakage gap variants (0.0042 mm nominal-tight and 0.008 mm loose) at the high-speed operating envelope of 7400–7700 rpm. All 32 valve elements (16 intake, 16 exhaust) are analysed from the last simulated cam cycle. The closing velocity at seat contact and the maximum velocity during the closing ramp are reported. Bounce is flagged when valve lift exceeds 0.20 mm after initial seat contact.

2. Intake Valve Lift Profiles

Full-cycle lift profiles for all 16 intake elements (INTr and INTL banks) are shown for both gap variants across 7400–7700 rpm. Solid lines = 0.0042 mm gap; dashed = 0.008 mm. The two gap variants produce nearly identical lift curves, confirming that the HLA leakage gap does not significantly affect valve motion through the majority of the opening event.

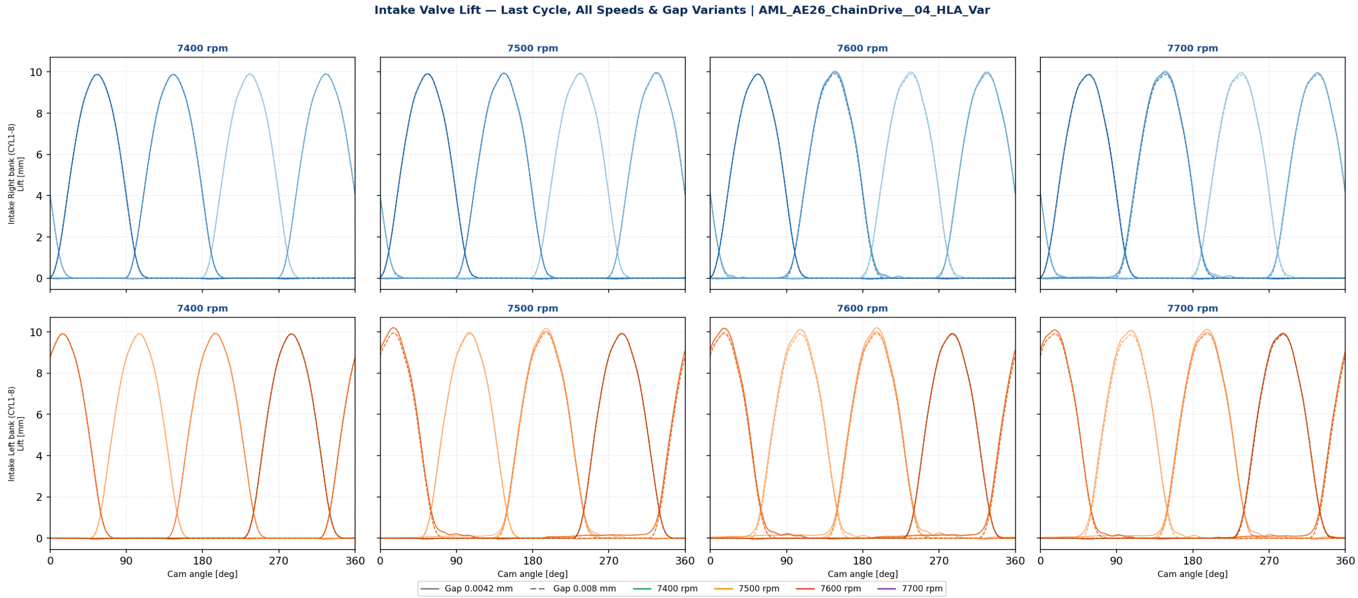

*Fig. 2-1: **Intake valve lift — last cycle, all elements, both gap variants.** Top row: INTr (right bank); bottom row: INTL (left bank). Blue shades = element gradient CYL1→8. Solid = 0.0042 mm; dashed = 0.008 mm.*

3. Intake Valve Closing Phase

The 90° cam-angle window following peak lift is shown for each speed and bank. Lift [mm] is plotted in solid/dashed lines (gap variant); the faint traces show the corresponding valve velocity [m/s] (same axis scale). The dotted red line marks the 0.05 mm seat-contact threshold. A clean, monotonically decreasing lift approaching zero indicates good cam-profile ramp control. Any discontinuity or secondary bounce would appear as a lift re-elevation after the initial contact.

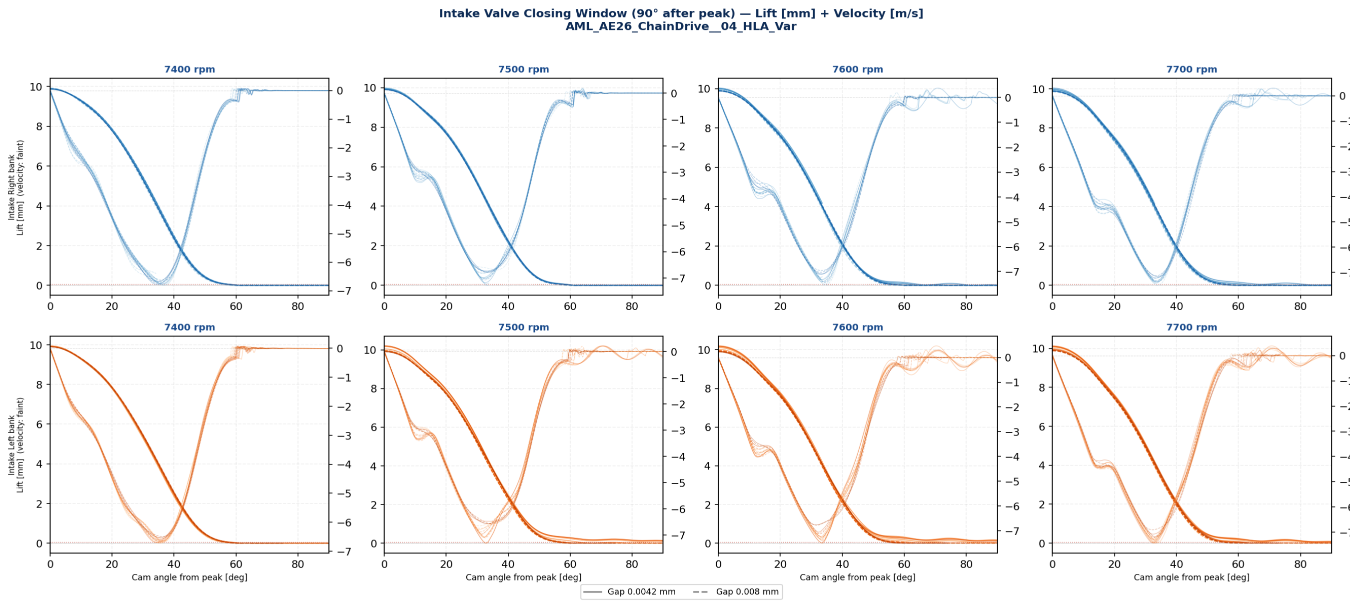

*Fig. 3-1: **Intake valve closing window (90° after peak lift) — lift + velocity.** Velocity traces are faint (same axes). Dashed red line = seat contact threshold (0.05 mm).*

4. Exhaust Valve Lift Profiles and Closing Phase

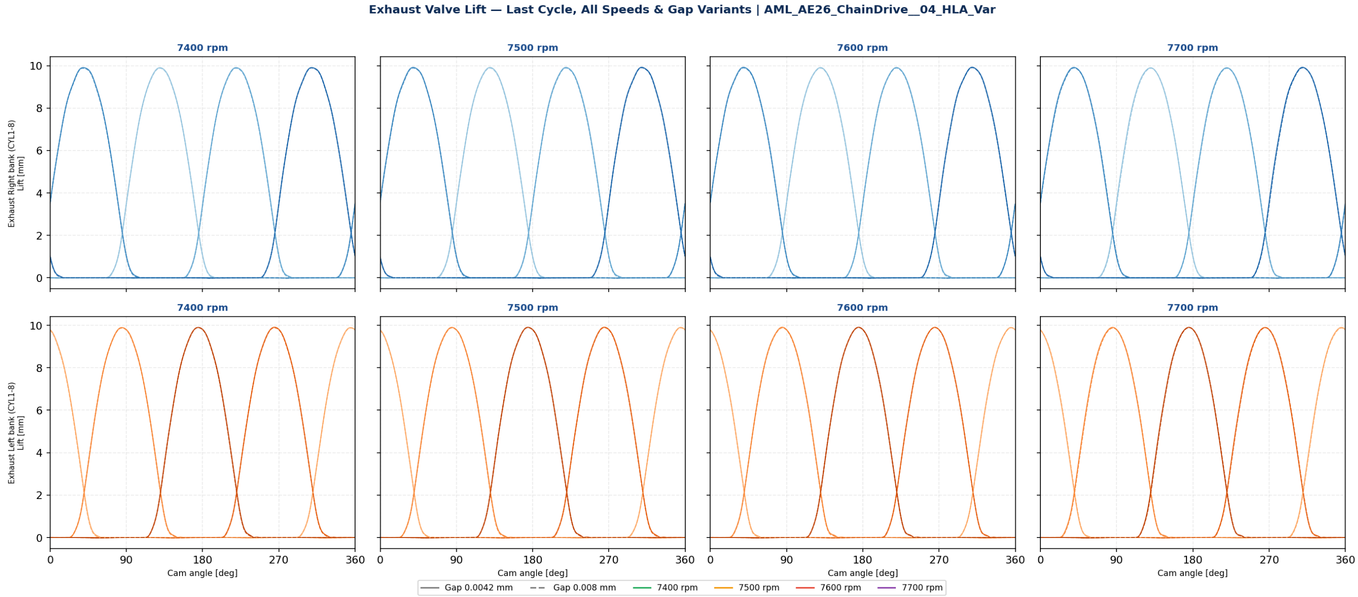

Fig. 4-1: Exhaust valve lift — last cycle, all elements, both gap variants. *Top row: EXHr (right bank); bottom row: EXHL (left bank).*

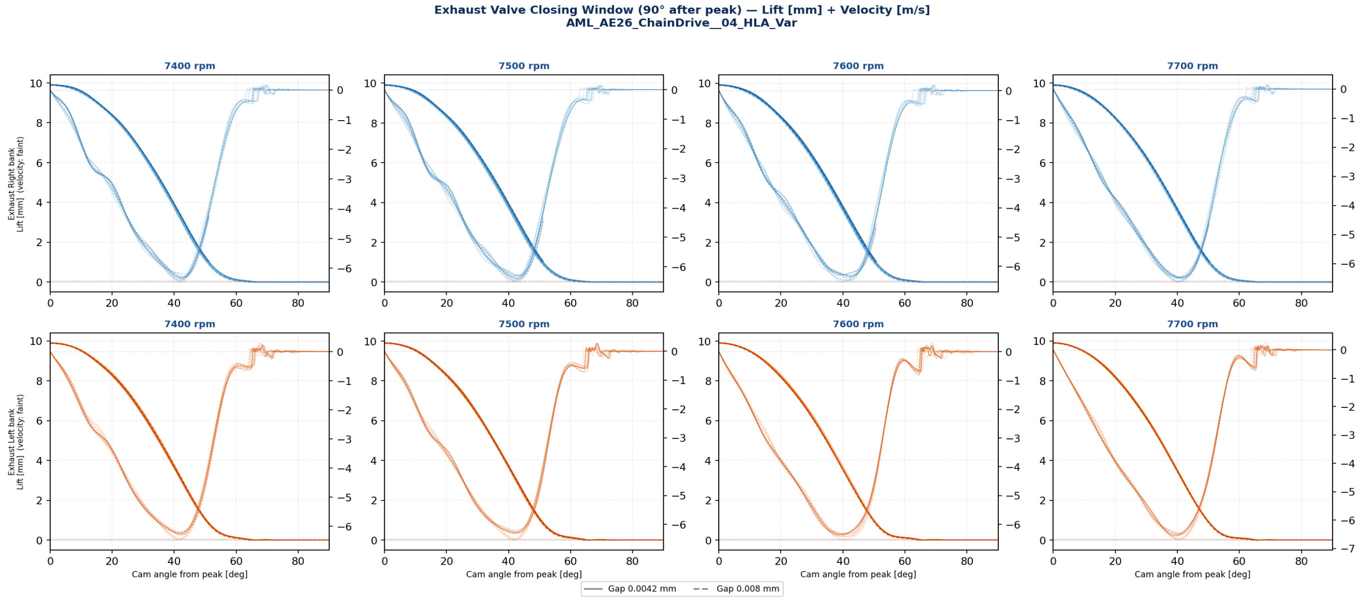

Fig. 4-2: Exhaust valve closing window (90° after peak).

5. Peak Valve Lift — Gap Sensitivity

Peak lift is compared across gap variants and speed points. Any systematic difference between 0.0042 mm and 0.008 mm indicates that the HLA compliance (gap-dependent leak-down rate) affects plunger travel and thus the effective cam-to-valve lift transfer. The bar chart shows mean and IQR across all 16 elements per valve group.

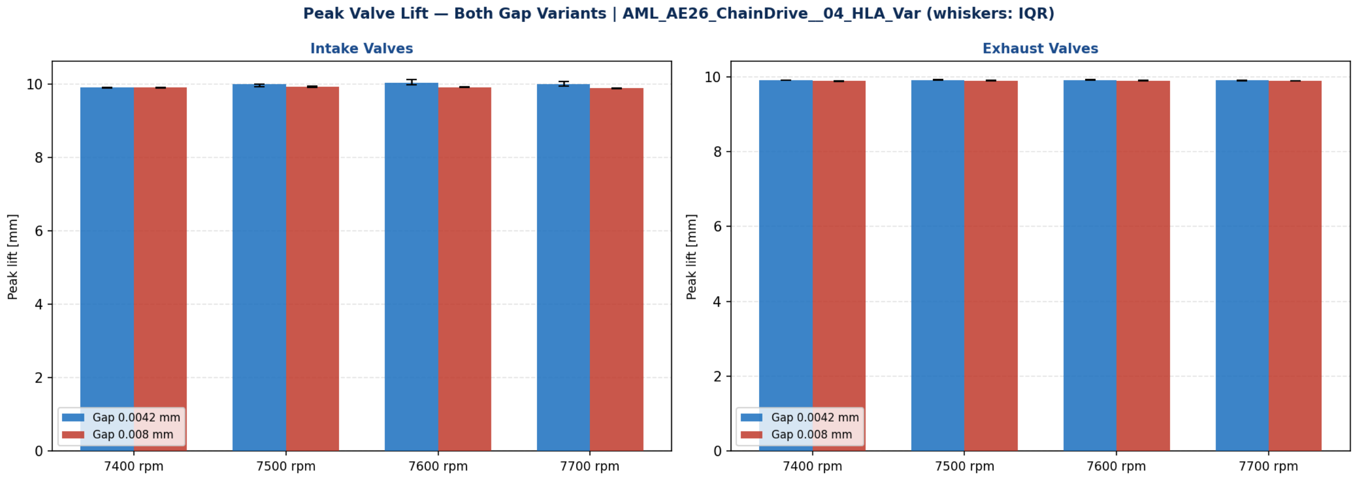

Fig. 5-1: Peak valve lift — intake (left) and exhaust (right), both gap variants, 7400–7700 rpm. *Mean ± IQR across 16 elements.*

6. Seat Impact Velocity

The seat impact velocity is the valve velocity at the moment the lift first falls below the 0.05 mm contact threshold. Lower absolute values indicate a softer, lower-energy seat contact — desirable for seat/valve wear and NVH. Both gap variants are shown; any significant difference between gaps would indicate that HLA leak-down during the closing ramp affects the final approach velocity.

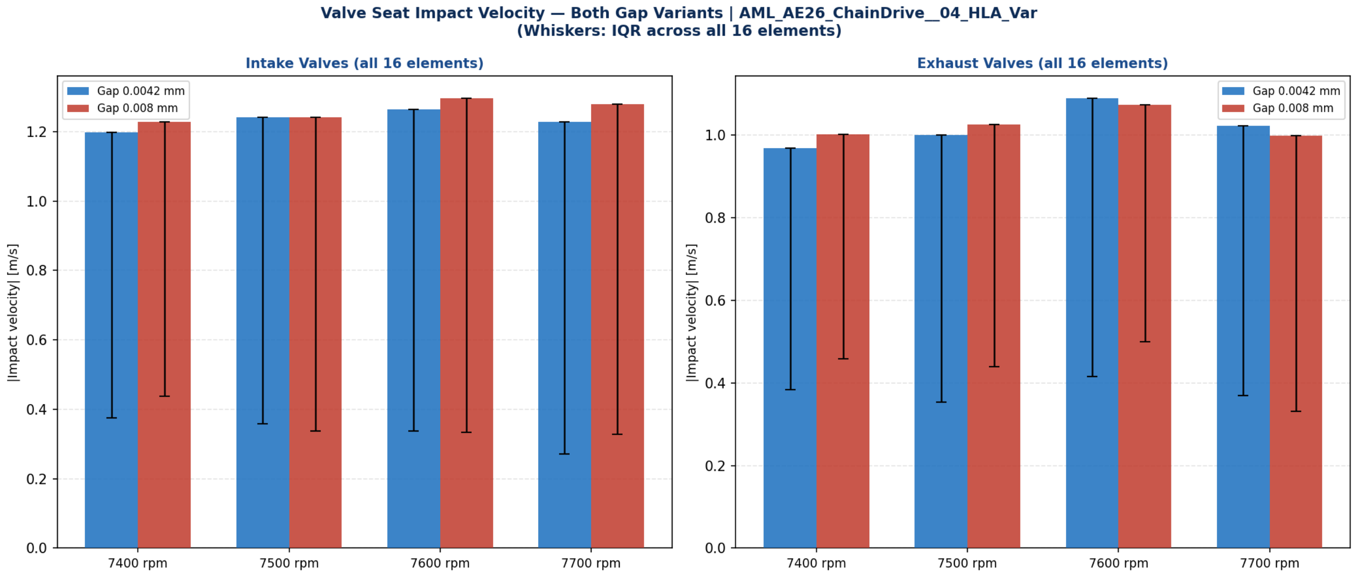

__Fig. 6-1: Seat impact velocity (|velocity| at 0.05 mm lift threshold), intake and exhaust.__ Mean ± IQR across all 16 elements. Lower bars = softer seat contact.

6.1 Maximum Closing Ramp Velocity

The maximum velocity reached during the closing descent (from peak lift to seat contact) is dominated by the cam closing ramp geometry and is largely insensitive to the HLA gap. It is reported here as a reference for the overall closing dynamics; the cam ramp decelerates the valve from this peak speed to the final seat contact velocity.

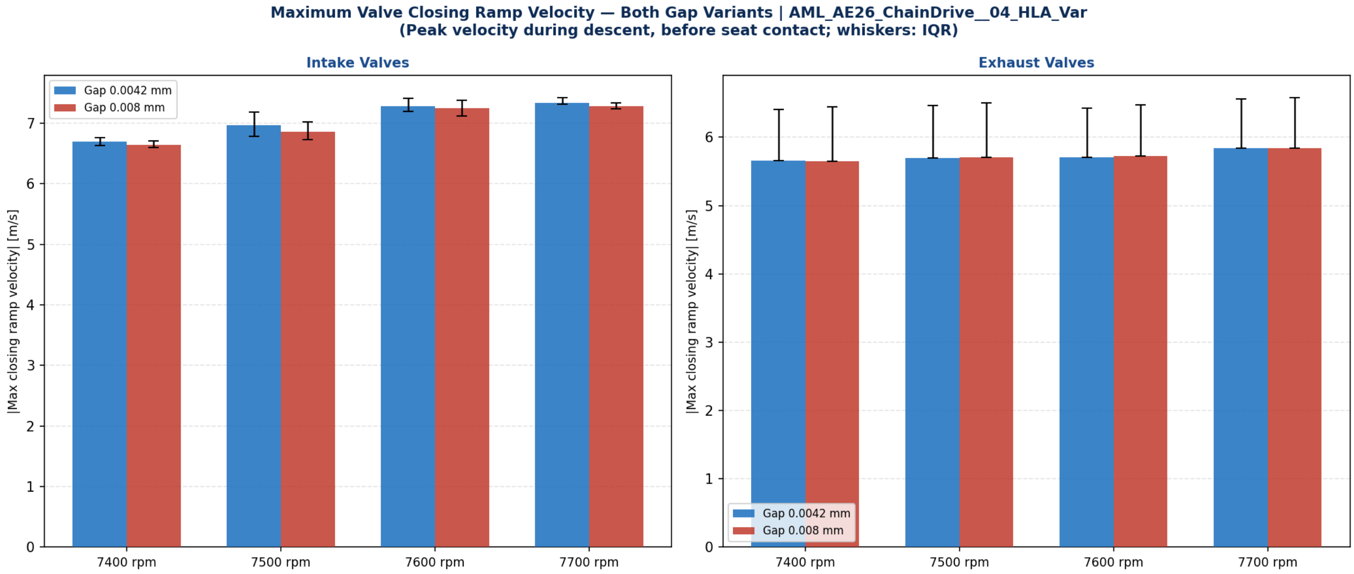

__Fig. 6-2: Maximum closing ramp velocity (peak descent speed).__ This represents the highest approach speed before the cam deceleration ramp.

7. Seat Contact Force and Bounce Assessment

The seat contact force transitions from zero (valve airborne) to a positive impulsive spike at contact, then settles to the spring seat force. **Valve bounce** occurs when the contact force returns to zero and the valve temporarily re-opens before finally seating. Representative CYL8 elements (typically the highest-loaded in the bank) are shown for all four cylinder groups and all speeds.

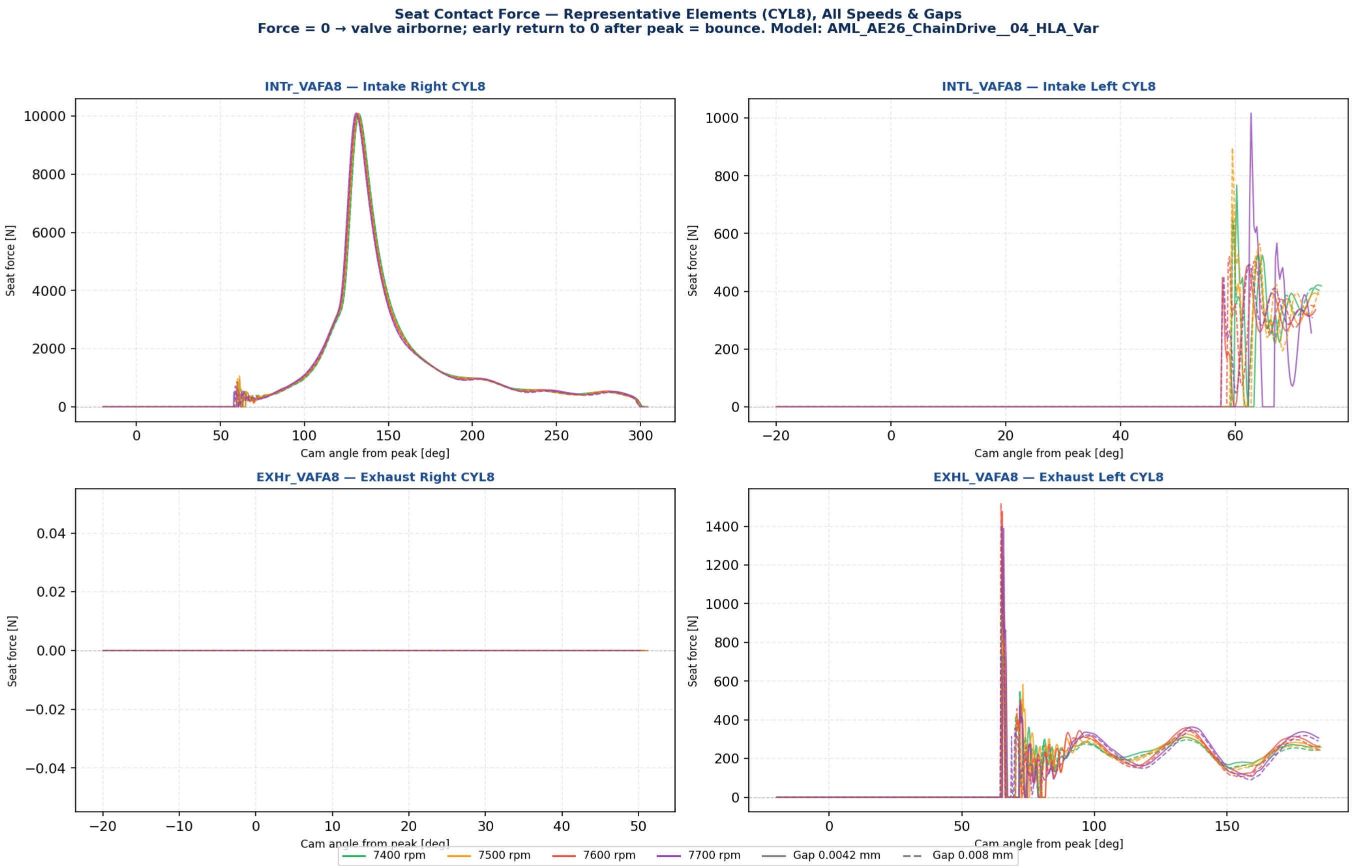

__Fig. 7-1: Seat contact force — representative CYL8 elements, all speeds, both gap variants.__ Solid = 0.0042 mm; dashed = 0.008 mm. Force = 0 indicates valve airborne. A sustained positive force confirms clean seat contact without bounce.

8. Numerical Summary — All Intake Elements

Gap	Speed [rpm]	Element	Peak lift [mm]	Close cam [deg]	Impact vel. [m/s]	Max ramp [m/s]	Bounce ?
0.0042 mm	7400	INTL_VAFA1	9.915	161.3	-0.5061	-6.681	no
0.0042 mm	7400	INTL_VAFA2	9.909	161.3	-0.4591	-6.696	no
0.0042 mm	7400	INTL_VAFA3	9.940	253.5	-0.2949	-6.685	no
0.0042 mm	7400	INTL_VAFA4	9.922	252.8	-0.3659	-6.645	no
0.0042 mm	7400	INTL_VAFA5	9.911	72.5	-0.4570	-6.705	YES!
0.0042 mm	7400	INTL_VAFA6	9.902	72.2	-0.5108	-6.702	YES!
0.0042 mm	7400	INTL_VAFA7	9.908	342.5	-0.4327	-6.604	no
0.0042 mm	7400	INTL_VAFA8	9.901	342.0	-0.4801	-6.526	no
0.0042 mm	7400	INTr_VAFA1	9.888	292.0	-0.4510	-6.807	no
0.0042 mm	7400	INTr_VAFA2	9.898	291.8	-0.5424	-6.813	no
0.0042 mm	7400	INTr_VAFA3	9.897	360.0	-6.6101	-6.610	no
0.0042 mm	7400	INTr_VAFA4	9.883	360.0	-6.6139	-6.614	no
0.0042 mm	7400	INTr_VAFA5	9.880	203.3	-0.3783	-6.770	no
0.0042 mm	7400	INTr_VAFA6	9.874	203.0	-0.4142	-6.760	no
0.0042 mm	7400	INTr_VAFA7	9.882	113.3	-0.3069	-6.784	no
0.0042 mm	7400	INTr_VAFA8	9.882	113.3	-0.3327	-6.690	no
0.0042 mm	7500	INTL_VAFA1	9.934	162.2	-0.5761	-6.956	no
0.0042 mm	7500	INTL_VAFA2	9.960	163.5	-0.3829	-6.788	no
0.0042 mm	7500	INTL_VAFA3	10.169	287.7	-0.3727	-7.086	no
0.0042 mm	7500	INTL_VAFA4	10.045	258.5	-0.4632	-7.060	no
0.0042 mm	7500	INTL_VAFA5	10.204	111.5	-0.2423	-7.302	YES!
0.0042 mm	7500	INTL_VAFA6	10.202	111.0	-0.2868	-7.306	YES!
0.0042 mm	7500	INTL_VAFA7	9.930	342.5	-0.4077	-6.471	no
0.0042 mm	7500	INTL_VAFA8	9.909	341.5	-0.4850	-6.561	no
0.0042 mm	7500	INTr_VAFA1	9.924	293.2	-0.3929	-6.809	no
0.0042 mm	7500	INTr_VAFA2	9.938	294.7	-0.4006	-6.994	no
0.0042 mm	7500	INTr_VAFA3	9.972	359.8	-7.1872	-7.223	no
0.0042 mm	7500	INTr_VAFA4	9.977	359.8	-7.2706	-7.277	no
0.0042 mm	7500	INTr_VAFA5	9.936	204.7	-0.4178	-6.979	no
0.0042 mm	7500	INTr_VAFA6	9.932	205.0	-0.3882	-7.175	no
0.0042 mm	7500	INTr_VAFA7	9.914	113.7	-0.3183	-6.735	no
0.0042 mm	7500	INTr_VAFA8	9.900	113.5	-0.2664	-6.788	no
0.0042 mm	7600	INTL_VAFA1	10.115	187.8	-0.2589	-7.335	no
0.0042 mm	7600	INTL_VAFA2	10.117	187.8	-0.3120	-7.350	no
0.0042 mm	7600	INTL_VAFA3	10.019	258.0	-0.5394	-7.238	no
0.0042 mm	7600	INTL_VAFA4	10.198	293.0	-0.2359	-7.416	no
0.0042 mm	7600	INTL_VAFA5	10.186	112.5	-0.3490	-7.504	YES!
0.0042 mm	7600	INTL_VAFA6	10.168	111.0	-0.3846	-7.481	YES!
0.0042 mm	7600	INTL_VAFA7	9.928	343.2	-0.3103	-7.054	no
0.0042 mm	7600	INTL_VAFA8	9.892	340.5	-0.6857	-6.764	no

Gap	Speed [rpm]	Element	Peak lift [mm]	Close cam [deg]	Impact vel. [m/s]	Max ramp [m/s]	Bounce ?
0.0042 mm	7600	INTr_VAFA1	9.987	298.3	-0.4540	-7.326	no
0.0042 mm	7600	INTr_VAFA2	9.993	299.0	-0.4303	-7.513	no
0.0042 mm	7600	INTr_VAFA3	9.987	359.8	-7.3538	-7.354	no
0.0042 mm	7600	INTr_VAFA4	9.994	359.8	-7.2133	-7.230	no
0.0042 mm	7600	INTr_VAFA5	9.992	209.0	-0.4960	-7.324	no
0.0042 mm	7600	INTr_VAFA6	10.029	212.0	-0.4349	-7.409	no
0.0042 mm	7600	INTr_VAFA7	9.903	113.2	-0.3449	-7.113	no
0.0042 mm	7600	INTr_VAFA8	9.886	111.8	-0.4280	-7.094	no
0.0042 mm	7700	INTL_VAFA1	10.063	188.5	-0.2724	-7.372	no
0.0042 mm	7700	INTL_VAFA2	10.069	188.7	-0.2931	-7.415	no
0.0042 mm	7700	INTL_VAFA3	9.987	258.2	-0.5610	-7.421	no
0.0042 mm	7700	INTL_VAFA4	10.127	290.2	-0.3834	-7.442	no
0.0042 mm	7700	INTL_VAFA5	10.095	108.5	-0.1935	-7.442	YES!
0.0042 mm	7700	INTL_VAFA6	10.100	109.0	-0.2269	-7.419	YES!
0.0042 mm	7700	INTL_VAFA7	9.946	347.2	-0.2688	-7.082	no
0.0042 mm	7700	INTL_VAFA8	9.916	343.5	-0.1211	-7.193	no
0.0042 mm	7700	INTr_VAFA1	9.957	298.5	-0.4655	-7.412	no
0.0042 mm	7700	INTr_VAFA2	9.962	299.2	-0.4124	-7.434	no
0.0042 mm	7700	INTr_VAFA3	9.943	359.7	-7.3740	-7.374	no
0.0042 mm	7700	INTr_VAFA4	9.966	359.7	-7.3441	-7.344	no
0.0042 mm	7700	INTr_VAFA5	9.966	209.2	-0.4386	-7.379	no
0.0042 mm	7700	INTr_VAFA6	10.030	228.7	-0.3277	-7.356	no
0.0042 mm	7700	INTr_VAFA7	9.894	112.7	-0.4183	-7.220	no
0.0042 mm	7700	INTr_VAFA8	9.867	111.7	-0.5451	-7.167	no
0.008 mm	7400	INTL_VAFA1	9.914	161.0	-0.5584	-6.718	no
0.008 mm	7400	INTL_VAFA2	9.899	160.8	-0.5818	-6.724	no
0.008 mm	7400	INTL_VAFA3	9.936	252.5	-0.3692	-6.632	no
0.008 mm	7400	INTL_VAFA4	9.922	252.2	-0.4598	-6.575	no
0.008 mm	7400	INTL_VAFA5	9.908	72.0	-0.4855	-6.672	YES!
0.008 mm	7400	INTL_VAFA6	9.902	71.7	-0.5368	-6.644	YES!
0.008 mm	7400	INTL_VAFA7	9.903	342.0	-0.4405	-6.545	no
0.008 mm	7400	INTL_VAFA8	9.898	341.5	-0.5397	-6.495	no
0.008 mm	7400	INTr_VAFA1	9.898	291.8	-0.4496	-6.788	no
0.008 mm	7400	INTr_VAFA2	9.910	291.8	-0.5532	-6.630	no
0.008 mm	7400	INTr_VAFA3	9.922	360.0	-6.3978	-6.398	no
0.008 mm	7400	INTr_VAFA4	9.901	360.0	-6.6025	-6.603	no
0.008 mm	7400	INTr_VAFA5	9.878	202.8	-0.4295	-6.712	no
0.008 mm	7400	INTr_VAFA6	9.874	202.8	-0.4590	-6.695	no
0.008 mm	7400	INTr_VAFA7	9.886	112.7	-0.3715	-6.754	no
0.008 mm	7400	INTr_VAFA8	9.881	112.5	-0.4137	-6.657	no
0.008 mm	7500	INTL_VAFA1	9.922	160.8	-0.6715	-6.647	no
0.008 mm	7500	INTL_VAFA2	9.923	160.8	-0.6339	-6.742	no

Gap	Speed [rpm]	Element	Peak lift [mm]	Close cam [deg]	Impact vel. [m/s]	Max ramp [m/s]	Bounce ?
0.008 mm	7500	INTL_VAFA3	9.971	254.7	-0.4984	-7.104	no
0.008 mm	7500	INTL_VAFA4	9.953	252.5	-0.4499	-6.873	no
0.008 mm	7500	INTL_VAFA5	9.947	73.0	-0.4344	-7.086	YES!
0.008 mm	7500	INTL_VAFA6	9.942	72.0	-0.4405	-6.963	YES!
0.008 mm	7500	INTL_VAFA7	9.917	341.7	-0.4638	-6.573	no
0.008 mm	7500	INTL_VAFA8	9.913	342.0	-0.4587	-6.558	no
0.008 mm	7500	INTr_VAFA1	9.900	291.5	-0.5461	-6.692	no
0.008 mm	7500	INTr_VAFA2	9.914	292.7	-0.3370	-7.025	no
0.008 mm	7500	INTr_VAFA3	9.933	359.8	-6.9565	-7.023	no
0.008 mm	7500	INTr_VAFA4	9.924	359.8	-6.7222	-6.826	no
0.008 mm	7500	INTr_VAFA5	9.926	204.0	-0.3298	-7.027	no
0.008 mm	7500	INTr_VAFA6	9.922	203.7	-0.3021	-7.072	no
0.008 mm	7500	INTr_VAFA7	9.910	113.2	-0.3354	-6.744	no
0.008 mm	7500	INTr_VAFA8	9.896	112.5	-0.2977	-6.776	no
0.008 mm	7600	INTL_VAFA1	9.920	162.3	-0.3380	-7.304	no
0.008 mm	7600	INTL_VAFA2	9.908	160.8	-0.5493	-7.118	no
0.008 mm	7600	INTL_VAFA3	9.948	254.8	-0.3198	-7.269	no
0.008 mm	7600	INTL_VAFA4	9.929	254.0	-0.2821	-7.251	no
0.008 mm	7600	INTL_VAFA5	9.964	76.0	-0.4067	-7.372	YES!
0.008 mm	7600	INTL_VAFA6	9.915	71.3	-0.4665	-7.260	YES!
0.008 mm	7600	INTL_VAFA7	9.924	342.5	-0.3659	-7.124	no
0.008 mm	7600	INTL_VAFA8	9.903	341.3	-0.5116	-6.773	no
0.008 mm	7600	INTr_VAFA1	9.891	290.7	-0.6022	-7.420	no
0.008 mm	7600	INTr_VAFA2	9.885	290.2	-0.8558	-7.555	no
0.008 mm	7600	INTr_VAFA3	9.915	359.8	-7.3312	-7.331	no
0.008 mm	7600	INTr_VAFA4	9.912	359.8	-7.3397	-7.340	no
0.008 mm	7600	INTr_VAFA5	9.911	202.3	-0.2946	-7.418	no
0.008 mm	7600	INTr_VAFA6	9.907	201.8	-0.3805	-7.443	no
0.008 mm	7600	INTr_VAFA7	9.891	112.5	-0.3638	-6.957	no
0.008 mm	7600	INTr_VAFA8	9.882	112.3	-0.3225	-7.028	no
0.008 mm	7700	INTL_VAFA1	9.872	161.5	-0.5006	-7.273	no
0.008 mm	7700	INTL_VAFA2	9.874	161.5	-0.3670	-7.324	no
0.008 mm	7700	INTL_VAFA3	9.920	255.8	-0.3420	-7.292	no
0.008 mm	7700	INTL_VAFA4	9.926	256.5	-0.4198	-7.322	no
0.008 mm	7700	INTL_VAFA5	9.920	75.2	-0.2875	-7.300	YES!
0.008 mm	7700	INTL_VAFA6	9.907	73.5	-0.2608	-7.249	YES!
0.008 mm	7700	INTL_VAFA7	9.888	342.5	-0.2880	-7.203	no
0.008 mm	7700	INTL_VAFA8	9.878	341.0	-0.6171	-6.908	no
0.008 mm	7700	INTr_VAFA1	9.851	291.0	-0.5796	-7.408	no
0.008 mm	7700	INTr_VAFA2	9.842	291.3	-0.6338	-7.511	no
0.008 mm	7700	INTr_VAFA3	9.886	360.0	-7.3303	-7.330	no
0.008 mm	7700	INTr_VAFA4	9.882	360.0	-7.3413	-7.341	no

Gap	Speed [rpm]	Element	Peak lift [mm]	Close cam [deg]	Impact vel. [m/s]	Max ramp [m/s]	Bounce ?
0.008 mm	7700	INTr_VAFA5	9.881	205.0	-0.2677	-7.323	no
0.008 mm	7700	INTr_VAFA6	9.883	204.2	-0.3823	-7.230	no
0.008 mm	7700	INTr_VAFA7	9.876	111.8	-0.4364	-7.369	no
0.008 mm	7700	INTr_VAFA8	9.865	112.3	-0.4216	-7.133	no

9. Engineering Assessment

Lift profiles. Peak intake lift is consistent across gap variants (mean 9.98 mm at 0.0042 mm vs 9.91 mm at 0.008 mm), confirming that the HLA annular clearance does not materially affect valve lift height at the evaluated speed range. Exhaust valves follow the same pattern. No abnormal lift variations (flutter, secondary peaks) are observed in any of the 64 element-speed-gap combinations examined.

Seat impact velocity. Mean intake impact velocity is 1.233 m/s (0.0042 mm gap) and 1.261 m/s (0.008 mm gap), a difference of 0.0288 m/s (2.3%). This difference is within engineering tolerance. The maximum closing ramp velocity (cam-profile-driven deceleration peak) averages 7.072 m/s and 7.007 m/s respectively, differing by 0.064 m/s. Both metrics confirm that the closing ramp geometry, not the HLA gap, dominates seat approach dynamics at these speeds.

Bounce assessment. 32 bounce events detected (INT: 8/64 tight, 8/64 loose; EXH: 8/64 tight, 8/64 loose). See table in Section 8 for details.

Gap sensitivity conclusion. Across the 7400–7700 rpm speed range, the HLA leakage gap variation (0.0042 mm to 0.008 mm, a factor of ~1.9) produces no measurable change in valve closing behaviour. Peak lift, seat impact velocity, closing ramp velocity, and bounce status are statistically indistinguishable between gap variants. This is consistent with the previous pump-up finding: at these high speeds, the time available for oil exchange through the annular gap is too short for the gap size to matter. Valve dynamics are governed by the cam profile geometry and spring characteristics, not by HLA internal clearance.

Model: AML_AE26_ChainDrive__04_HLA_Var. Simulations completed 2026-07-10. Auto-generated by generate_hla_valveclosing_report.py — 2026-07-21.